E - You WILL Like Sigma Problem

分值：450 分

Problem Statement

> Your lucky Greek letter for today is sigma. Solve this problem that uses sigma twice, and good fortune will surely come your way.
>
> 你今日的幸运希腊字母是 sigma（Σ）。解决这个用到了两次 sigma 的问题，好运一定会降临到你身上。

You are given a sequence of positive integers $A = (A_1, \cdots, A_N)$ of length N and a sequence of positive integers $B = (B_1, \cdots, B_M)$ of length M.

给定长度为 N 的正整数序列 $A = (A_1, \cdots, A_N)$，以及长度为 M 的正整数序列 $B = (B_1, \cdots, B_M)$。

Find the value, modulo 998244353, of $\sum_{i=1}^{N} \sum_{j=1}^{M} A_i \cdot B_j \cdot (i \bmod j)$.

求 $\sum_{i=1}^{N} \sum_{j=1}^{M} A_i \cdot B_j \cdot (i \bmod j)$ 的值模 998244353 的结果。

Constraints

- $1 \le N, M \le 5 \times 10^5$
- $1 \le A_i, B_j \le 5 \times 10^5$
- All input values are integers.
- $1 \le N, M \le 5 \times 10^5$
- $1 \le A_i, B_j \le 5 \times 10^5$
- 所有输入值均为整数。

Input

The input is given from Standard Input in the following format:

输入按照以下格式从标准输入给出：

$N\ M$
$A_1\ A_2\ \cdots\ A_N$
$B_1\ B_2\ \cdots\ B_M$

Output

Output the answer on a single line.

将答案输出在一行内。

Sample Input 1

```
6 4
1 6 9 2 3 1
1 10 3 7
```

Sample Output 1

```
508
```

The sum of the following 24 values is 508.

以下 24 个值的和为 508。

- $A_1 \cdot B_1 \cdot (1 \bmod 1) = 1 \cdot 1 \cdot 0 = 0$
- $A_1 \cdot B_2 \cdot (1 \bmod 2) = 1 \cdot 10 \cdot 1 = 10$
- $A_1 \cdot B_3 \cdot (1 \bmod 3) = 1 \cdot 3 \cdot 1 = 3$
- $A_1 \cdot B_4 \cdot (1 \bmod 4) = 1 \cdot 7 \cdot 1 = 7$
- $A_2 \cdot B_1 \cdot (2 \bmod 1) = 6 \cdot 1 \cdot 0 = 0$
- $A_2 \cdot B_2 \cdot (2 \bmod 2) = 6 \cdot 10 \cdot 0 = 0$
- $A_2 \cdot B_3 \cdot (2 \bmod 3) = 6 \cdot 3 \cdot 2 = 36$

- $A_2 \cdot B_4 \cdot (2 \bmod 4) = 6 \cdot 7 \cdot 2 = 84$
- $A_3 \cdot B_1 \cdot (3 \bmod 1) = 9 \cdot 1 \cdot 0 = 0$
- $A_3 \cdot B_2 \cdot (3 \bmod 2) = 9 \cdot 10 \cdot 1 = 90$
- $A_3 \cdot B_3 \cdot (3 \bmod 3) = 9 \cdot 3 \cdot 0 = 0$
- $A_3 \cdot B_4 \cdot (3 \bmod 4) = 9 \cdot 7 \cdot 3 = 189$
- $A_4 \cdot B_1 \cdot (4 \bmod 1) = 2 \cdot 1 \cdot 0 = 0$
- $A_4 \cdot B_2 \cdot (4 \bmod 2) = 2 \cdot 10 \cdot 0 = 0$
- $A_4 \cdot B_3 \cdot (4 \bmod 3) = 2 \cdot 3 \cdot 1 = 6$
- $A_4 \cdot B_4 \cdot (4 \bmod 4) = 2 \cdot 7 \cdot 0 = 0$
- $A_5 \cdot B_1 \cdot (5 \bmod 1) = 3 \cdot 1 \cdot 0 = 0$
- $A_5 \cdot B_2 \cdot (5 \bmod 2) = 3 \cdot 10 \cdot 1 = 30$
- $A_5 \cdot B_3 \cdot (5 \bmod 3) = 3 \cdot 3 \cdot 2 = 18$
- $A_5 \cdot B_4 \cdot (5 \bmod 4) = 3 \cdot 7 \cdot 1 = 21$
- $A_6 \cdot B_1 \cdot (6 \bmod 1) = 1 \cdot 1 \cdot 0 = 0$
- $A_6 \cdot B_2 \cdot (6 \bmod 2) = 1 \cdot 10 \cdot 0 = 0$
- $A_6 \cdot B_3 \cdot (6 \bmod 3) = 1 \cdot 3 \cdot 0 = 0$
- $A_6 \cdot B_4 \cdot (6 \bmod 4) = 1 \cdot 7 \cdot 2 = 14$
- $A_1 \cdot B_1 \cdot (1 \bmod 1) = 1 \cdot 1 \cdot 0 = 0$
- $A_1 \cdot B_2 \cdot (1 \bmod 2) = 1 \cdot 10 \cdot 1 = 10$
- $A_1 \cdot B_3 \cdot (1 \bmod 3) = 1 \cdot 3 \cdot 1 = 3$
- $A_1 \cdot B_4 \cdot (1 \bmod 4) = 1 \cdot 7 \cdot 1 = 7$
- $A_2 \cdot B_1 \cdot (2 \bmod 1) = 6 \cdot 1 \cdot 0 = 0$
- $A_2 \cdot B_2 \cdot (2 \bmod 2) = 6 \cdot 10 \cdot 0 = 0$
- $A_2 \cdot B_3 \cdot (2 \bmod 3) = 6 \cdot 3 \cdot 2 = 36$
- $A_2 \cdot B_4 \cdot (2 \bmod 4) = 6 \cdot 7 \cdot 2 = 84$
- $A_3 \cdot B_1 \cdot (3 \bmod 1) = 9 \cdot 1 \cdot 0 = 0$

- $A_3 \cdot B_2 \cdot (3 \bmod 2) = 9 \cdot 10 \cdot 1 = 90$
- $A_3 \cdot B_3 \cdot (3 \bmod 3) = 9 \cdot 3 \cdot 0 = 0$
- $A_3 \cdot B_4 \cdot (3 \bmod 4) = 9 \cdot 7 \cdot 3 = 189$
- $A_4 \cdot B_1 \cdot (4 \bmod 1) = 2 \cdot 1 \cdot 0 = 0$
- $A_4 \cdot B_2 \cdot (4 \bmod 2) = 2 \cdot 10 \cdot 0 = 0$
- $A_4 \cdot B_3 \cdot (4 \bmod 3) = 2 \cdot 3 \cdot 1 = 6$
- $A_4 \cdot B_4 \cdot (4 \bmod 4) = 2 \cdot 7 \cdot 0 = 0$
- $A_5 \cdot B_1 \cdot (5 \bmod 1) = 3 \cdot 1 \cdot 0 = 0$
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- $A_6 \cdot B_1 \cdot (6 \bmod 1) = 1 \cdot 1 \cdot 0 = 0$
- $A_6 \cdot B_2 \cdot (6 \bmod 2) = 1 \cdot 10 \cdot 0 = 0$
- $A_6 \cdot B_3 \cdot (6 \bmod 3) = 1 \cdot 3 \cdot 0 = 0$
- $A_6 \cdot B_4 \cdot (6 \bmod 4) = 1 \cdot 7 \cdot 2 = 14$

Sample Input 2

```
20 20
36625 195265 98908 111868 111868 47382 147644 472464 472464 416653 111868 195265 327
327972 111868 416653 177330 340019 262769 47382 262769 47382 340019 47382 262769 327
```

Sample Output 2

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58141644
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